

Immifin Architecture

Version: 1.0

Last Updated: June 2026

Vision

Immifin is an AI-powered immigration platform that helps employment-based immigrants understand Visa Bulletin movement, track priority dates, receive alerts, and interact with an AI immigration assistant.

Primary goals:

- Accurate immigration data
 - Fast website
 - Secure authentication
 - Subscription-based SaaS
 - AI-powered assistance
 - Low operating cost
 - Easy deployment
 - Easy scaling
-

Technology Stack

Frontend

- Next.js (App Router)
- React
- TypeScript
- TailwindCSS

Hosting:

- Cloudflare Pages
-

Backend

- Next.js API Routes

Future:

- Cloudflare Workers (if needed)
-

Database

Development

- Supabase (Immifin Dev)

Production

- Supabase (Immifin Production)

Database stores:

- Profiles
 - Immigration profiles
 - Subscriptions
 - Audit logs
 - Future user-generated data
-

Authentication

Provider:

Clerk

Development:

- Clerk Development Instance

Production:

- Clerk Production Instance

Identity Source:

- Clerk User ID

Database Mapping:

profiles.clerk_user_id

Payments

Provider:

Stripe

Development:

- Stripe Test Mode

Production:

- Stripe Live Mode

Plans

- Free
 - Basic
 - Pro
-

AI

Primary

- OpenAI GPT

Future

- Multiple model support

Possible providers

- OpenAI
- Anthropic
- Google Gemini

Architecture should allow switching models.

Environments

Development

Cloudflare

Preview deployment

Clerk

Development

Supabase

Immifin Dev

Stripe

Test Mode

Production

Cloudflare

Production deployment

Clerk

Production

Supabase

Immifin Production

Stripe

Live

Deployment Flow

GitHub

↓

Cursor

↓

Push to main



Cloudflare Build



Preview Deployment



Testing



Production Deployment

User Roles

Guest

Anonymous visitor

Permissions

- Read public Visa Bulletin
- View movement tracker
- View history

Cannot

- Save profile
 - Receive alerts
 - AI assistant
-

Free

Registered user

Can

- Save immigration profile
 - Monthly email alerts
 - Dashboard
-

Basic

Everything in Free

Plus

- Priority alerts
 - Better analytics
 - Faster notifications
-

Pro

Everything in Basic

Plus

- AI Immigration Assistant
 - WhatsApp chatbot
 - iMessage support (future)
 - Google Messages support (future)
 - Advanced reports
-

Admin

Full platform management

Can

- Refresh Visa Bulletin
 - Manage users
 - View subscriptions
 - View audit logs
 - Manage announcements
-

Data Sources

Primary

Google Sheets

Published CSV

Website always reads from published CSV.

Admin manually refreshes data.

Future

USCIS

DOS

Other immigration datasets

Admin Console

Future pages

Dashboard

Users

Subscriptions

Visa Bulletin

Refresh Data

Audit Logs

System Health

Security

Authentication

Clerk

Authorization

Role-based

Secrets

Never committed to Git

Cloudflare Environment Variables

Supabase Service Role

Server only

Repository Structure

app/

components/

lib/

docs/

scripts/

supabase/

credentials/

Roadmap

Phase 1

- Authentication
 - User profiles
 - Admin
-

Phase 2

- Stripe
 - Free / Basic / Pro
-

Phase 3

- Admin Dashboard
-

Phase 4

- AI Assistant
-

Phase 5

- WhatsApp
 - SMS
 - Push Notifications
-

Design Principles

1. Keep infrastructure simple.
 2. Build for long-term maintainability.
 3. Never duplicate business logic.
 4. Prefer pull architecture over push.
 5. Cache aggressively.
 6. Keep monthly operating cost low.
 7. Security before convenience.
 8. Every feature must provide measurable value to immigrants.
-

Success Metrics

- Monthly Active Users
- Paid Subscribers
- Alert Delivery Success
- AI Satisfaction
- Page Load Speed
- Uptime
- Revenue
- Customer Retention